

Killian Gérard

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PROFILE

Mechatronics BEng graduate with two engineering internships in live energy operations — Vitol Group on UK gas turbines and VIVO Energy (Shell licensee) on LPG and oil infrastructure in Morocco. Hands-on experience building machine-learning models in Python (battery-life prediction and high-speed-rail vibration analysis) and end-to-end CAPEX/OPEX modelling and oil-demand forecasting from fieldwork. Trilingual: French, English and Spanish. Joining IE Madrid's MSc Fintech programme in September 2026.

PROFESSIONAL EXPERIENCE

JUL–AUG 2025

Engineering Internship

VIVO Energy (Shell Licensee) · Morocco

- Shadowed engineers across 3+ LPG and oil-infrastructure disciplines, supporting project development and management.
- Learned end-to-end CAPEX/OPEX modelling — Excel structure, oil-demand forecasting and tank-reserve analytics with fail-safe systems.
- Monitored oil-tank construction and welding under strict safety regulation; redesigned the division's engineer-onboarding deck.

JUL–AUG 2022

Engineering Shadowing

VPI Damhead Creek, Vitol Group · UK

- Shadowed real-time performance of Mitsubishi 701F3 gas turbines (~800 MW combined).
- Adjusted operational output against regional daily demand; supported procurement and formal reporting.

KEY PROJECTS

- 3RD YR ML for Remaining Useful Life of Li-ion batteries — LSTM on Stanford/MIT dataset, MAPE < 0.07%.
- 3RD YR Vibration analysis — high-speed-rail suspension concept with ML anomaly detection.
- 2ND YR Line-following robot firing ping-pong balls at a target after a set distance.
- 1ST YR IMechE robot car — drive into a wall and reverse precisely to start position.
- 1ST YR Scalable miniature wind-turbine aerofoil — JavaFoil, SolidWorks and wind-tunnel testing.

EDUCATION

Lancaster University

Mechatronics Engineering BEng (Hons), 2023–2026

University of Nottingham

Physics w/ Astronomy & Foundation, 2022–2023

Colfe's School, London

Private secondary & sixth form, 2015–2021

IE University, Madrid

MSc Fintech — from Sept 2026

LANGUAGES

French & English (native), Spanish (proficient), Arabic / Portuguese / Italian (elementary).

TECHNICAL SKILLS

Python (Pandas, NumPy, TensorFlow/Keras, scikit-learn, Matplotlib, Gradio), MATLAB, C/C++ embedded. Machine learning, time-series & predictive modelling. SolidWorks, AutoCAD, JavaFoil. CAPEX/OPEX modelling.

LEADERSHIP & AWARDS

EPALS Vice-President · RAF (CCF) Cadet, Corporal · Duke of Edinburgh's Award, Silver & Bronze.